CHIOMA OKEKE

602-740-7962 | okekegeraldenne@gmail.com | http://www.linkedin.com/in/okeke-chioma | https://github.com/Chioma-Okeke

EDUCATION

M.S. Computer Science, Arizona State University — Expected May 2027
Mastercard Foundation Scholar | Relevant Coursework: Advanced Operating Systems, Software Security

B.Eng. Computer Engineering, Covenant University — Aug 2022
CGPA: 4.67/5.0 | **Relevant Coursework:** Introduction to Neural Networks, Data Structures & Algorithms

SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL, Apex.

Backend: Node.js, Express.js, FastAPI, RESTful APIs, Salesforce Apex

Databases: PostgreSQL, MongoDB, Mongoose

Developer Tools: Git, Bitbucket, Linux, Kibana, Jest, Salesforce Code Analyzer

Concepts: Data Structures & Algorithms, OOP, Distributed Systems, CI/CD, Authentication & Authorization, RBAC, JWT, OAuth2

EXPERIENCE

EdPlus, Arizona State University - Software Development Assistant — Aug 2025 – Present

- Developed Apex-based backend automation for Salesforce record routing and user reassignment, handling insert and update events to maintain accurate record ownership as user availability changed.
- Expanded an Apex test optimization effort from a single assigned class to approximately **60%** of the existing test suite, raising overall code coverage to **85%** and strengthening release confidence.
- Developing an AI-assisted Salesforce code review workflow combining Salesforce Code Analyzer, custom deterministic checks, and AI analysis across seven quality, security, metadata, and release-readiness gates to improve review consistency and identify risks earlier.

Cavista Technologies- Software Engineer I | Healthcare Software — Mar 2023 – Aug 2025

- Collaborated with backend engineers to investigate production job failures using Kibana and code-level analysis, reducing investigation time by **35%** and accelerating fixes for recurring issues.
- Engineered a form-to-model workflow using Aurelia and TypeScript, transforming structured inputs into reusable data models for generating print-ready clinical documentation.
- Shipped a user-facing permissions feature in Aurelia and TypeScript that surfaced mobile-access information for applicable permissions, helping users understand the impact of permission changes before making them.

PROJECTS

Laundry Inventory Management System | Web Application

- Built a full-stack inventory and operations platform with RESTful services using Next.js, TypeScript, MongoDB, and Mongoose, centralizing customer, order, item-tracking, and business records across role-based workflows.
- Implemented JWT authentication, role-based authorization, and automated receipt and tag generation, reducing manual recordkeeping and improving staff efficiency by **40%**.

CarbonLint | Python Code Optimization Engine & Developer Infrastructure

- Built a Python code optimization engine that detects inefficient code, automatically applies performance improvements, and measures changes in runtime, memory consumption, and estimated CO_2 emissions.
- Architected a five-stage processing pipeline combining static analysis, AST-based transformations, and sandboxed benchmarking with restricted modules, memory limits, and execution timeouts.
- Exposed optimization capabilities through FastMCP and integrated a TypeScript VS Code extension using newline-delimited JSON over stdin/stdout; validated the system with 161 Python and TypeScript tests.

LLM Context Integrity & Prompt Injection Defense | Python / FastAPI / ML / MQTT

- Co-led research investigating prompt-injection and context-integrity attacks against LLM-enabled smart-home systems, evaluating direct, indirect, and tool-integrated attack scenarios.
- Evaluated a three-layer defense architecture combining embedding-based detection, input hardening, and deterministic output validation to study how layered controls reduce unsafe LLM-driven device actions.